Jacob Winefeld

jacob.winefeld@gmail.com • linkedin.com/in/jacobwinefeld • jacobwinefeld.github.io

Education

University of California, Los Angeles — Los Angeles, CA
B.S. & M.S. Mechanical Engineering, expected Jun 2028. GPA: 3.95, Dean's Honors, Dept Scholars Program — Sep 2024 - Present
Relevant Coursework: Feedback & Control Systems, Robotic Kinematics, Dynamical Systems, Mechanisms, Circuits

Experience

SpaceX — Redmond, WA
Mechanical Engineering Intern, Starlink Payloads — Jun 2026 - Sep 2026

- Designed a structural mechanism between the phased-array antenna and chassis of a future Starlink satellite to reduce launch loads and bending moments on antenna components while accommodating thermal expansion
- Derived component loads by hand from system-level coupled-load analysis (CLA); used Ansys beam and solid-element FEA to inform part design, weight reduction, rivet vs. bolt selection, and to report structural margins
- Coordinated with the RF team to assess how the design affects antenna performance and to identify needed testing; built and installed prototype hardware on a mock antenna, verifying designed degrees of freedom and tolerances
- Led qualification testing of a structural attachment on the V3 antenna: bounded worst-case launch and on-orbit loads & displacements, designed fixtures, and ran Instron failure tests across temperature extremes to report final margins

UCLA Biomechatronics Lab (PI: Prof. Veronica Santos) — Los Angeles, CA
Undergraduate Researcher — Jul 2025 - Present

- Built a 3-DoF robotic laryngoscope for autonomous vocal cord imaging from concept to cadaveric testing and FDA/IRB submission; designed the mechanism, sized motors, and wrote the calibration routine and safety limits
- Co-inventor on patent "*Robotic Platform for Flexible Endoscopy*"; $69K won at pitch competitions for startup launch
- Conducted hysteresis characterization of the tendon-driven endoscope tip using ArUco vision tracking; performed fatigue durability testing across 900+ cycles & force safety testing of all 3 DoFs using a load cell for FDA submission
- Designed housing, wiring, and illumination system for *Opti*, an open-source vision-based tactile sensor; policies trained on its optical flow, divergence, and curl signals resolved weight differences to 13.8g; co-authored paper in preparation
- Increased robotic hand bracket flexural load capacity by 75%, while improving serviceability & wire management
- Destructively tested new designs in PLA, PETG, PLA-CF, PA6-CF, determining optimal material choice & infill settings for the robotic hand, and developed lab-wide standards for 3D printing practices

Bruin Racing Baja SAE (collegiate off-road vehicle engineering competition) — Los Angeles, CA
Suspension Testing & Analysis Engineer — Sep 2026 - Present

- Building a data acquisition system with strain gauges on the tie rod and lower A-arm, shock potentiometers, and a chassis IMU, logged by an ESP32, to measure steering and suspension loads for FEA and shock tuning
- Developing a kinematic model that resolves suspension travel and vehicle acceleration, with corner mass, spring and damper data and geometry, into shock, A-arm and tie rod loads, to be validated against gauge data

Steering Project Engineer — Sep 2025 - Jun 2026

- Led the design and manufacturing of the team's first fully custom steel rack-and-pinion steering system, with 18% reduction in steering effort, improved lock-to-lock travel, serviceability, and zero failures across both competitions
- Conducted shear pin testing to quantify worst-case loads (~680 lb axial tie rod force), then performed hand calculations and FEA on rack bending to validate a 7/8" 4340 steel rack with ~40% yield safety margin
- Used spring scale measurements to drive FEA for pinion-on-rack gear tooth stresses and steering column torsion, leading to downsizing of column wall thickness; quantified 6° steering wheel backlash, stable across the full season

Skills

Software: ROS2, Python, Java, C/C++, MATLAB, Arduino, PID control, Git, NX, SolidWorks, Onshape, Fusion, Ansys (FEA)
Sensing & actuation: Dynamixel servos, ATI force/torque sensor, load cells, strain gauges, IMU, ArUco vision tracking, Instron
Manufacturing: TIG welding, CAM/CNC, manual machining, 3D printing, laser cutting, soldering, elastomer molding, GD&T, DFMA